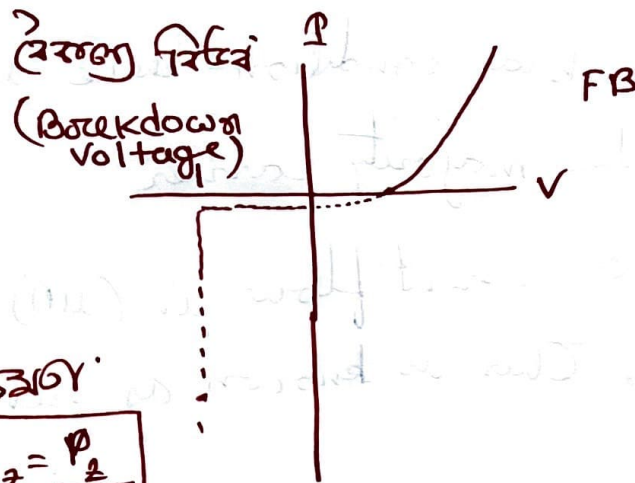


* Zener diode-



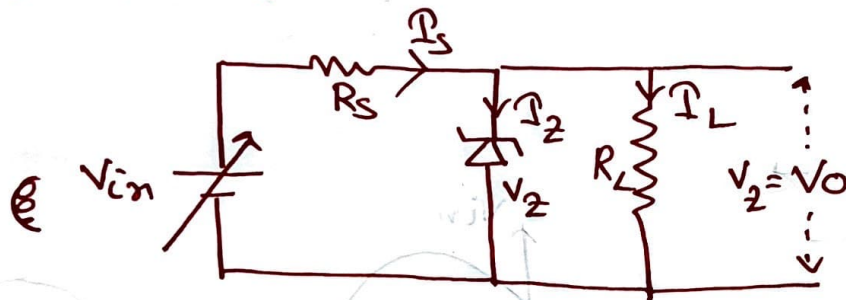
$V_z = \text{Breakdown Voltage}$

$P_z = \text{Power (अवस्थापन)}$

$$P_z = I_z V_z \therefore I_z = \frac{P_z}{V_z}$$

* Application-

① Voltage regulator (वोल्टेज रेगुलेटर)



$$① I_z = \frac{P_z}{V_z}$$

$$② I_L = \frac{V_z}{R_L}$$

$$③ I_s = I_L + I_z$$

$$④ V_{in} - V_z = I_s R_s \therefore R_s = \frac{V_{in} - V_z}{I_s}$$

* Breakdown

① Avalanche breakdown (अवस्थापन विच्छेद)

② Zener breakdown (जेनर विच्छेद)

$$* \frac{4.7V}{5.2V}$$

* The break down voltage ($V_Z = 5V$) and Power. ($P_Z = 100W$). If the input voltage ($V_{in} = 20V$) and load resistance ($R_L = 10\Omega$), then determine the value of R_S

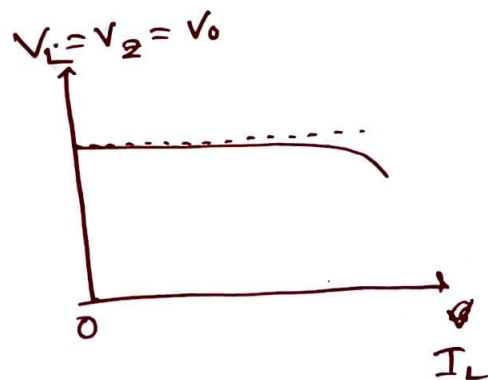
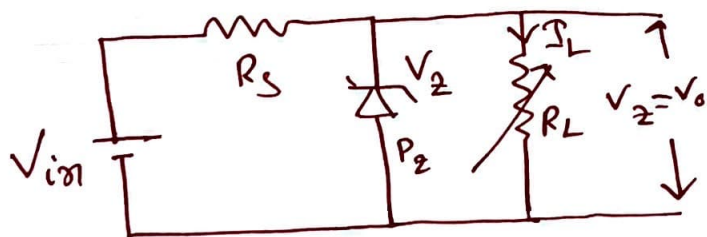
$$I_Z = \frac{P_Z}{V_Z} = \frac{100}{5} = 20 \text{ amp.}$$

$$I_L = \frac{V_Z}{R_L} = \frac{5}{10} = \frac{1}{2} \text{ amp.}$$

$$I_S = I_L + I_Z = 20 + \frac{1}{2} = \frac{41}{2} \text{ amp.}$$

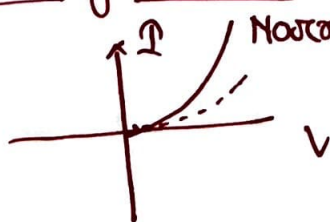
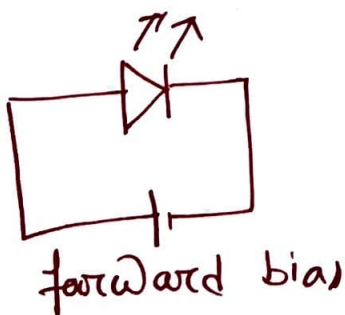
$$\therefore R_S = \frac{V_{in} - V_Z}{I_S} = \frac{20 - 5}{\frac{41}{2}} = \frac{30}{41} \Omega. (\text{Ans})$$

② Load regulator (ভোল্ট রেগুলেটর)



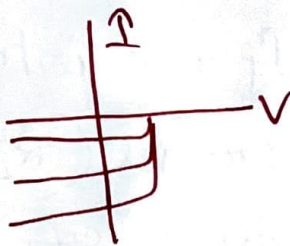
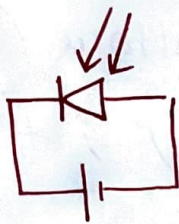
* Different types of diode -

① LED (Light Emitting diode)



Normal diode | No of carrier reduced due to recombination

* Photo diode -



* Solar cell

